

Final Review

Monday, Dec. 30. 4:30-6:30pm 3rd 207

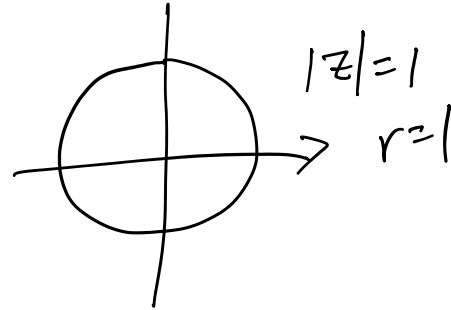
复数

$$z = x + iy, \quad re^{i\theta}$$

complex conjugate

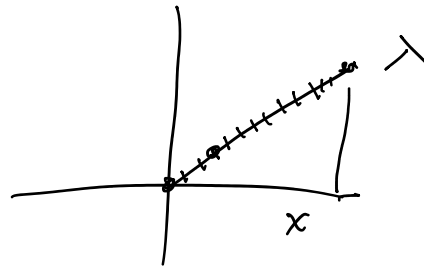
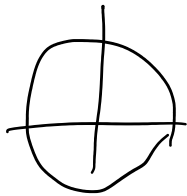
$$\bar{z} = x - iy, \quad re^{-i\theta}$$

$$\bar{z} = 1/z, \quad z^*$$



$$\frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_1}$$

euler formula



$$x + i\lambda x$$

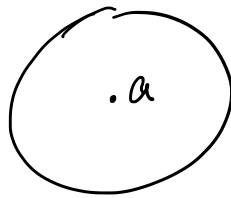
$$dz = dt + i\lambda dt$$

$$z = a + r \cos \theta + i(b + r \sin \theta)$$

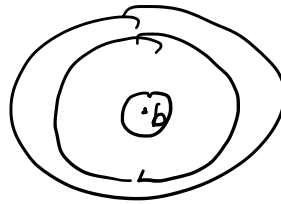
$$z = t + i\lambda t$$

$$\int_a^b f(z) (1 + i\lambda) dt$$

$$dz = \dots - \underline{d\theta}$$



Laurent



$$f(z) = \sum_{n=-\infty}^{+\infty} a_n (z-b)^n //$$

$$\underline{a_n} = \frac{1}{2\pi i} \oint \frac{f(s)}{(s-b)^{n+1}} ds$$

$$\frac{1}{(z-1)z^2}, \quad \underline{z=1}$$

$$(z-1)^{-1}$$

$$\underline{z^2} \cdot \underline{(z-1)^{-1}}$$

$$z^2 =$$

奇点: 可去奇点,

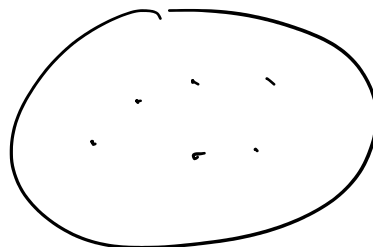
m-阶极点

本性奇点

$$\infty \quad z \rightarrow 1/z$$

留数定理

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^K \text{res} f(b_k)$$



a_{-1}

$$\lim_{z \rightarrow b} (z-b) f'(z)$$

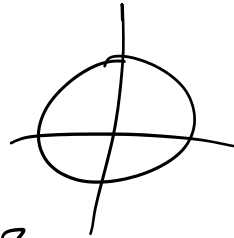
$$a_{-1} = \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} (z-b)^m f(z) \Big|_{z=b}$$

$$\int_0^{2\pi} R(\sin \theta, \cos \theta) d\theta$$

$$\sin \theta = \frac{z^2 - 1}{2iz}$$

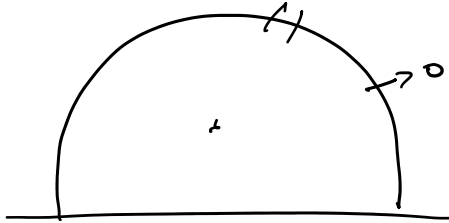
$$\cos \theta = \frac{z^2 + 1}{2z}$$

$$d\theta = \frac{dz}{iz}$$



$$\int_0^\pi \frac{1}{1 + \frac{1}{2} \cos \theta}$$

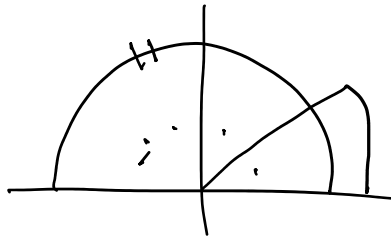
$$\int_{-\infty}^{+\infty} f(x) dx$$



$$\int_{-\infty}^{+\infty} \frac{dx}{(1+x^2)^3}$$

$$\int_{-\infty}^{+\infty} f(x) \cos x dx$$

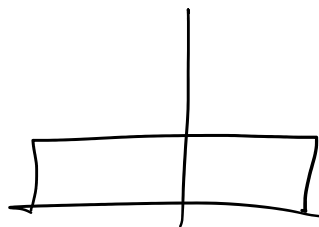
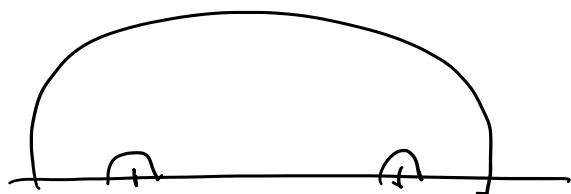
$$\int_C f(z) e^{ipz} dz$$



Jordan Theorem

$$\int_{C_R} g(z) e^{ipz} dz \rightarrow 0 \quad \text{as } R \rightarrow \infty$$

→ 0 - 3.2.30



$$u_{xx} + u_{yy} = 0$$

$$u_t - k(u_{xx} + u_{yy}) = 0$$

$$u_{tt} - a^2 u_{xx} = \underline{f(x, t)}$$

$$\underline{v(x, t)} \quad \downarrow \quad \underline{x^2} \quad x^2$$

$$\left\{ \begin{array}{ll} u|_{x=0} = 0 & u|_{x=l} = 0 \\ u|_{x=0} = 0 & u_x|_{x=l} = 0 \\ u_x|_{x=0} = 0 & u_x|_{x=l} = 0 \end{array} \right.$$

$$v = \underline{f(x)} \cdot \underline{\omega}$$

$$u_{xx} - 5u_{xy} + 6u_{yy} = 0 \quad x^2y$$

$$(D_x^2 - 5D_x D_y + 6D_y^2) u = 0$$

$$\phi_1(y+2x) + \phi_2(y+3x)$$

$$ax+by \quad \perp \quad ax+by$$

$$\frac{e^{ax+by}}{c} = \frac{1}{c} \cdot e^{ax+by}$$

$$c = L(a-b)$$

$$\cos(ax+by) \rightarrow$$

$$u_{xx} + u$$

$$u_t - a^2 u_{xx} = f(x, t)$$

Assume

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \underline{X_n(x)}$$

$$f(x, t) = \sum_{n=1}^{\infty} g_n(t) \underline{X_n(x)} \quad \checkmark$$

$$u|_{x=0} = u|_{x=l} = 0$$

$$\sum_{n=1}^{\infty} T_n''(t) X_n(x) - a^2 \sum_{n=1}^{\infty} T_n(t) \underline{X_n''(x)} = \sum_{n=1}^{\infty} g_n(t) X_n(x) \quad X_n(x)$$

$-\lambda X_n(x)$

$X_n(x)$: basis function
orthogonal

$n=2$

$$X_n''(x) = -\lambda X_n(x)$$

$$X_n(0) = X_n(l) = 0$$

$$T_n'(t) + a^2 \lambda T_n(t) = g_n(t)$$

~~$0 - Aw(1 - \frac{x}{l}) \cos \omega t$~~ $f(x, t)$

$$= \sum_{n=1}^{\infty} q_n \sin \frac{n\pi}{l} x$$

$$(1 - \frac{x}{l}) = \sum_{n=1}^{\infty} c_n \sin \frac{n\pi}{l} x \quad \cos \omega t$$

$q_n(t)$

$$\bar{n}=1$$

$$f(x,t) = e^{xt}, \quad \sin(xt)$$

$$g_n(t)$$